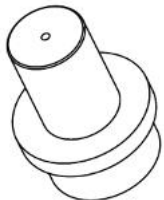


Sample Probe Position Adjustment

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1. Adjustment tool

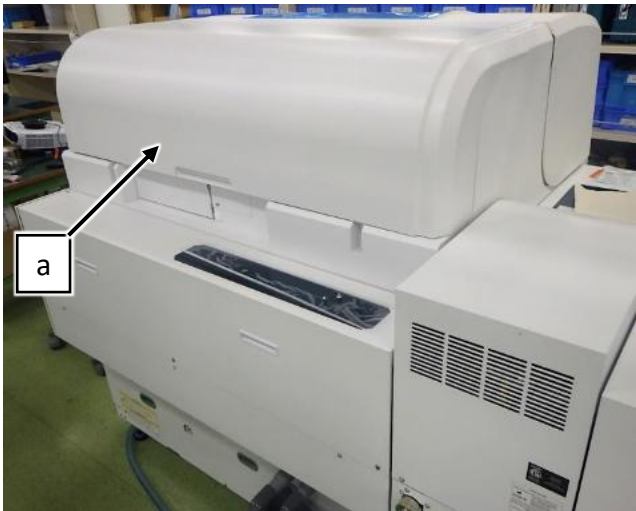
The following adjustment tools are used for this adjustment. Please refer to “S03-01 List of adjustment tools” for a list of jigs.

P/N	Name	Appearance
791-1830	CELL CENTER ADJUSTING TOOL	
792-6427	SAMPLE PROBE ADJUSTING TOOL 2	 X2

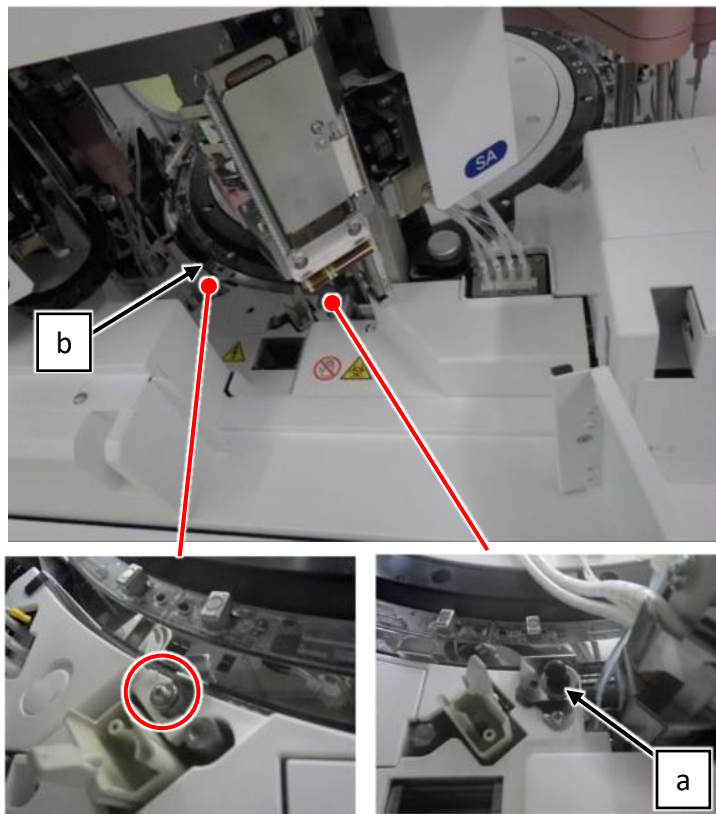
2. Preparation

2.1 Removing the covers

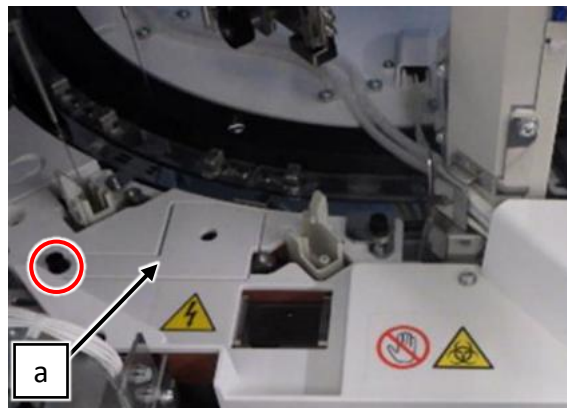
- (1) Open TOP COVER R ASSY (a).



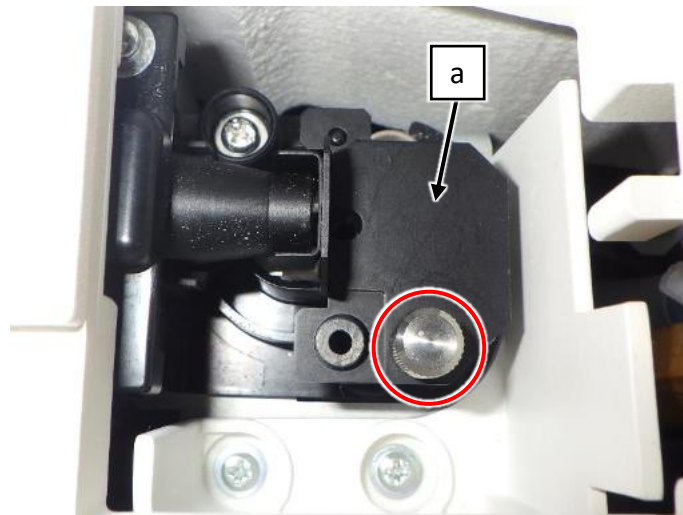
- (2) Pull the latch (a) and loose the screw, and then remove Cell Cover (b)



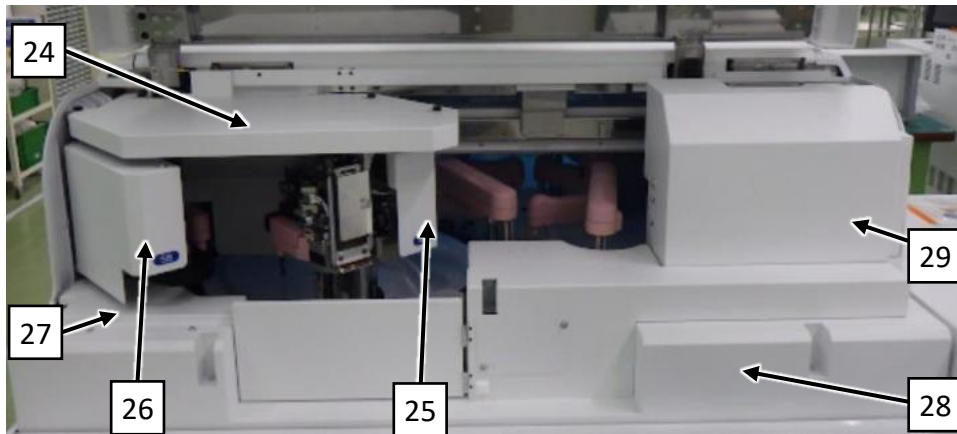
- (3) Pull the latch and remove SWS COVER U ASSY (a).



- (4) Remove the thumb screw and COVER SWS T (a).



- (5) Remove No.24, No.27 and No.28 covers (refer to “S06-41 Removal and attachment of covers”).



- (6) Loosen the screw and remove COVER MSW ASSY (a).

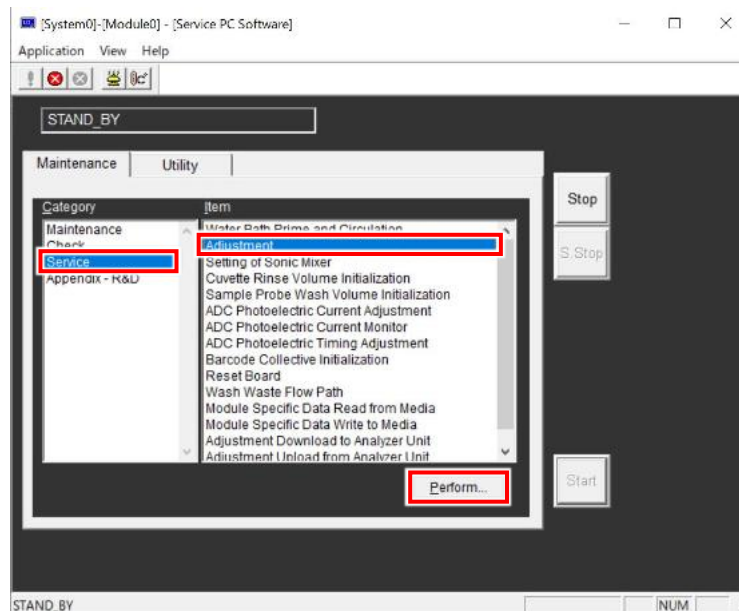


- (7) Turn on Maintenance switch.



2.2 Start SSW

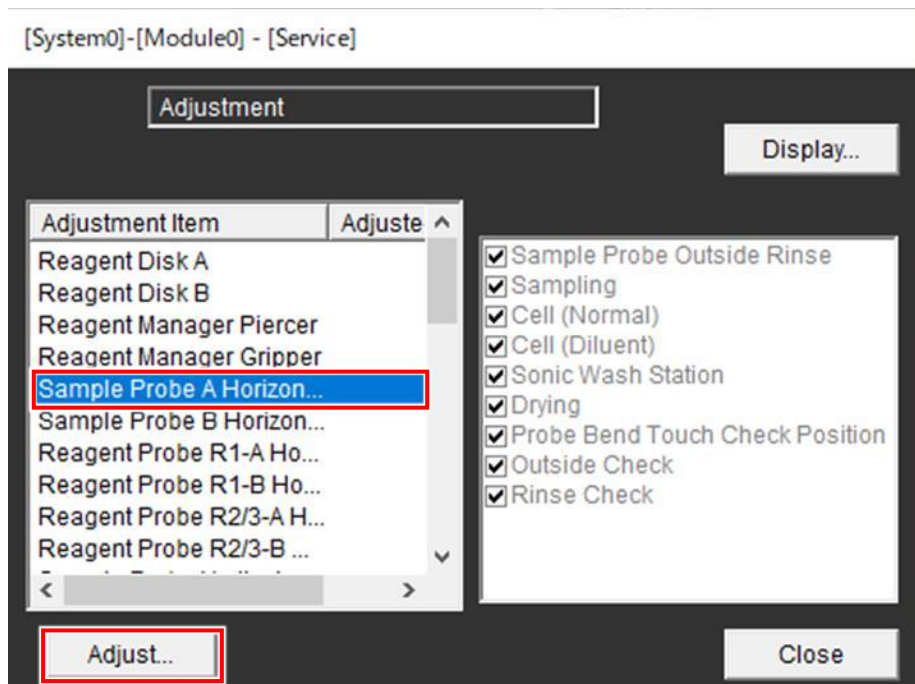
- (1) Refer to “17 pro v2 SSW Adjustment Procedure”.
- (2) Select [Service] > [Adjustment] > [Perform...] in SSW.



3. Adjustment of Sample Probe A Horizontal

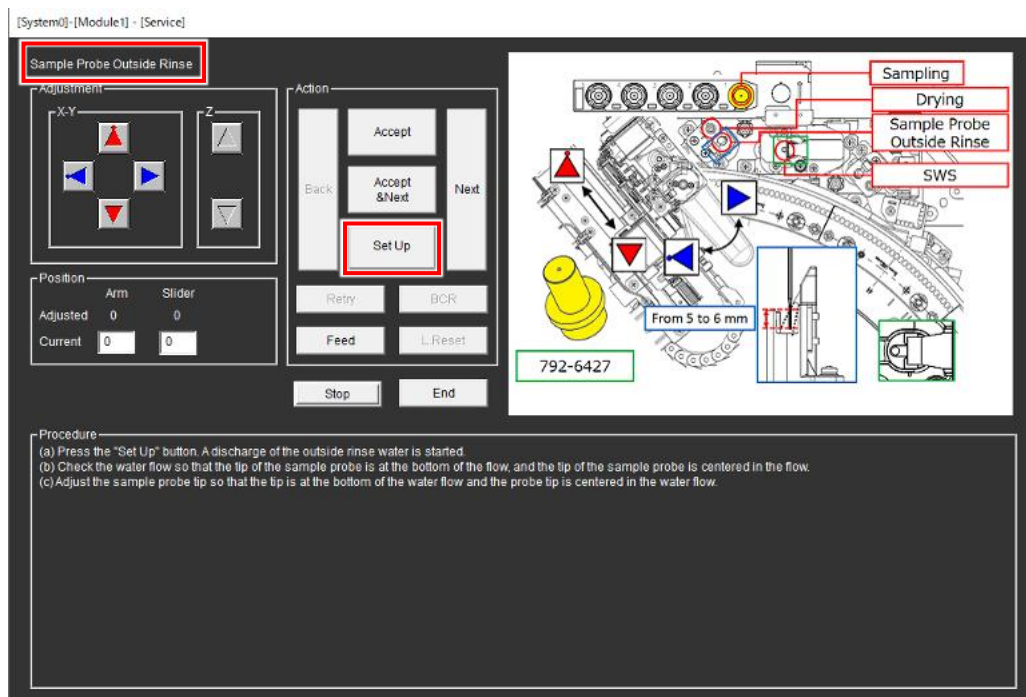
The adjustment procedure for SB is basically same as the procedure for SA.

- (1) Select [Sample Probe A Horizontal] from the Adjustment item and select [Adjust...] in SSW.



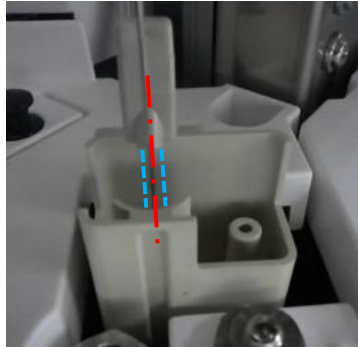
3.1 Sample Probe Outside Rinse position

- (1) Make sure that the adjustment subject is "Sample Probe Outside Rinse" at the upper left of SSW. If not, select [Next] repeatedly until "Sample Probe Outside Rinse" appears.
- (2) Select [Set Up].



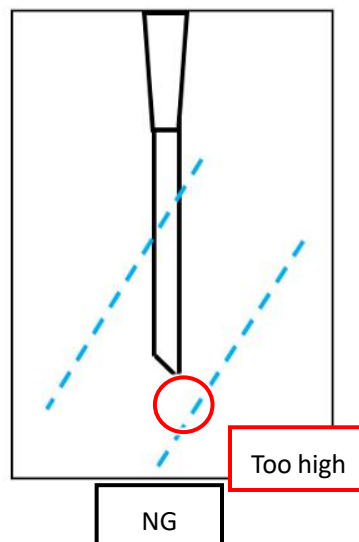
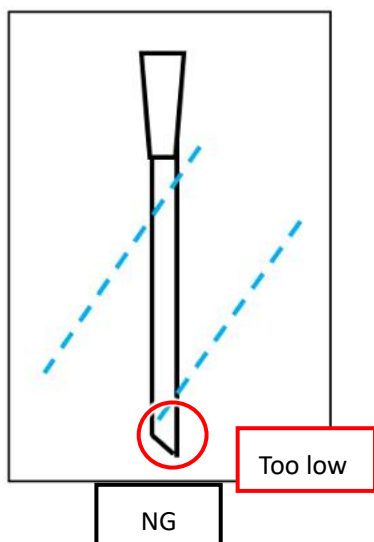
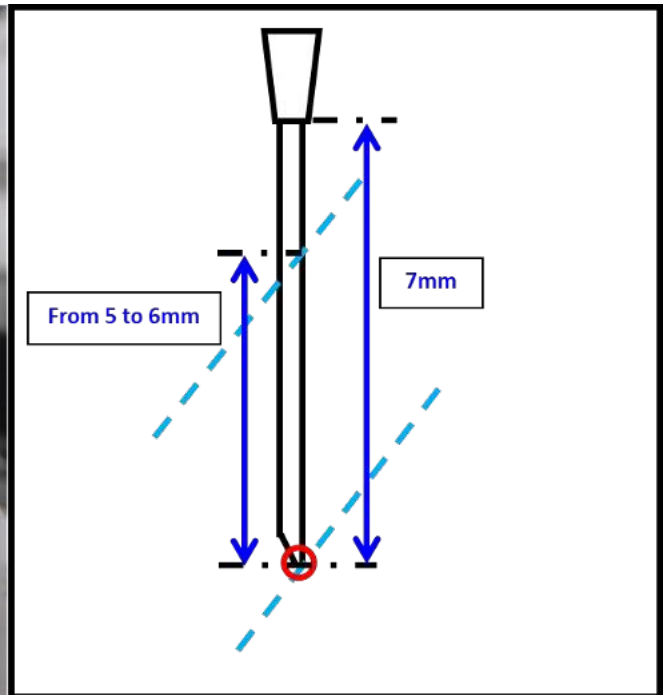
- (3) Make sure that the tip of the sample probe is positioned at the center of the outside wash flow at the washing station.

If not, perform Sample Probe Wash Volume Adjustment (refer to “S03-09 Sample Probe Wash Volume Adjustment”).

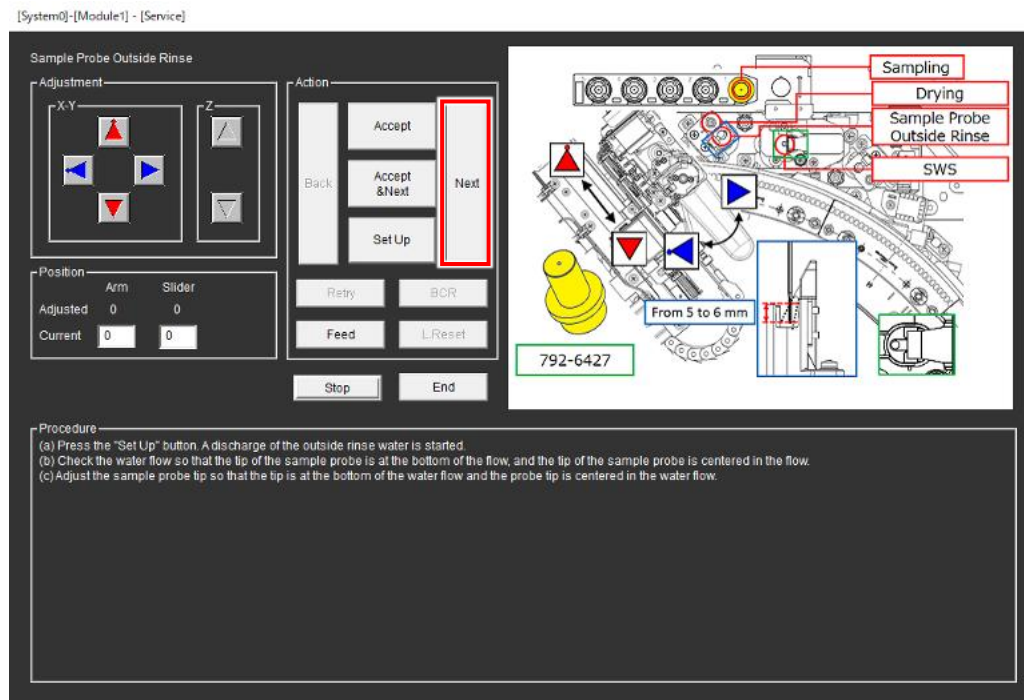


- (4) Make sure that the tip of the sample probe is positioned in the bottom of the outside wash flow.

If not, perform Sample Probe Wash Volume Adjustment (refer to “S03-09 Sample Probe Wash Volume Adjustment”).



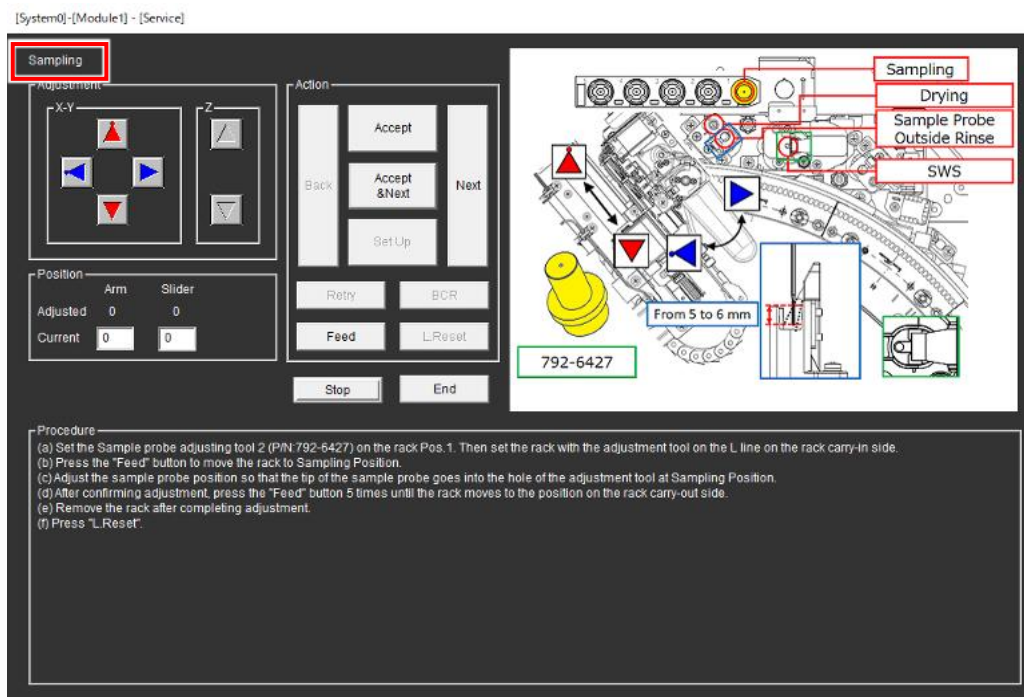
(5) Select [Next].



3.2 Sampling position

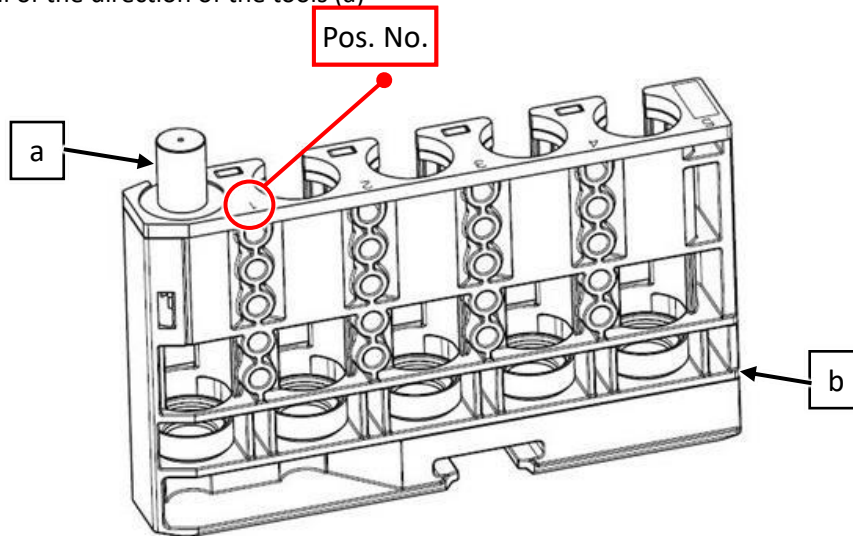
(1) Make sure that the adjustment subject is "Sampling" at the upper left of SSW.

If not, select [Next] repeatedly until "Sampling" appears.



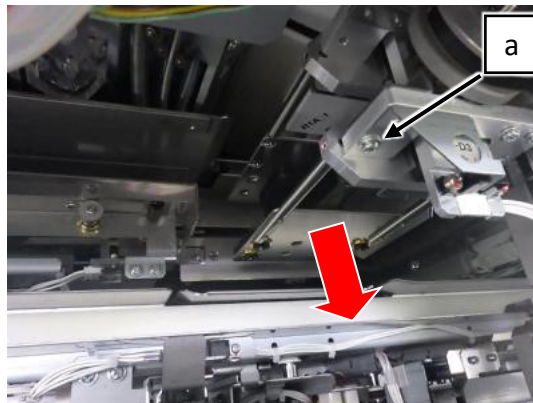
- (2) Set the tool (a) on Pos.1 of the rack (b).

Note: Be careful of the direction of the tools (a)

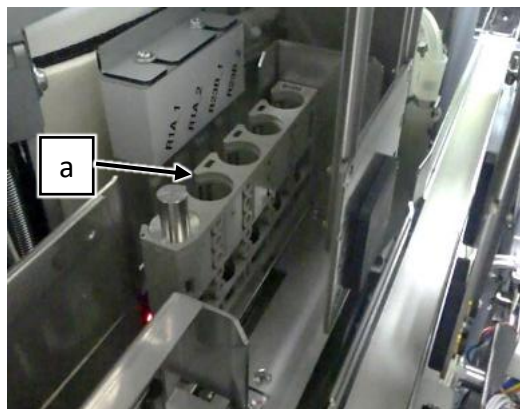


(a): SAMPLE PROBE ADJUSTING TOOL 2 (P/N:792-6427)

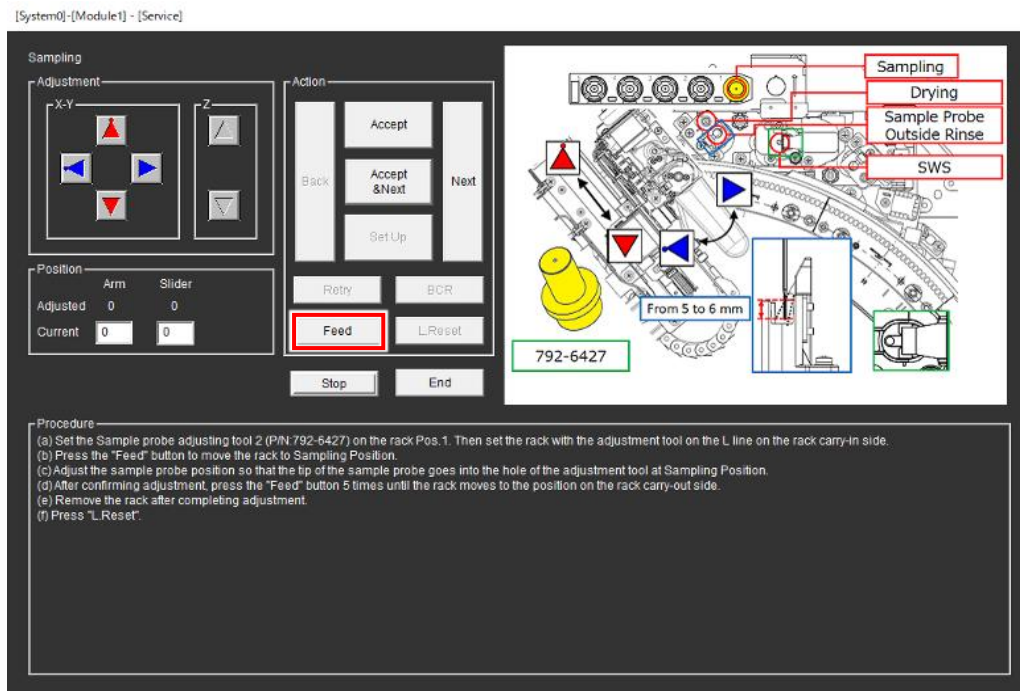
- (3) Move Rack handle (a) as shown below.



- (4) Set the rack (a) on the right end of Sampling line.



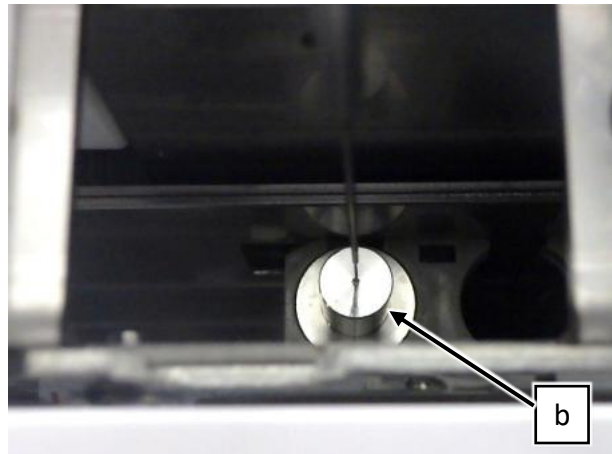
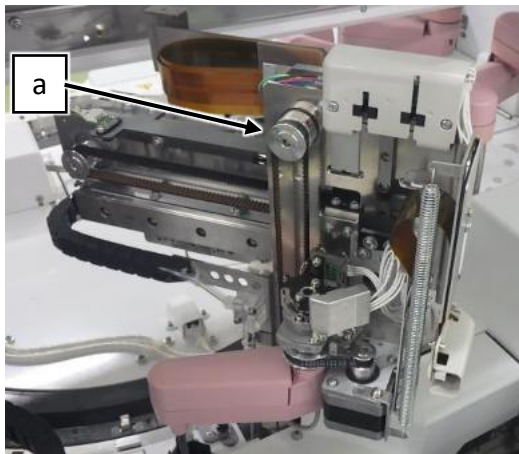
(5) Select [Feed].



(6) Lower Sample probe by rotating the pulley (a) and make sure that the tip of Sample probe is at the center of the tool (b).

If it is at the center, proceed to the step (9).

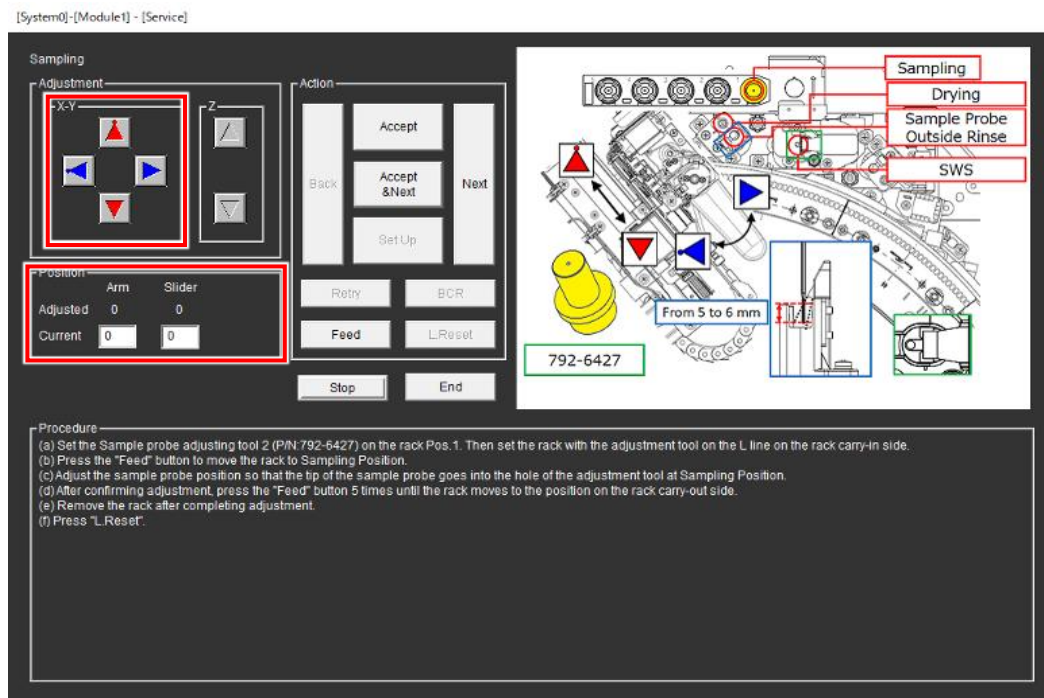
If not, proceed to the next step.



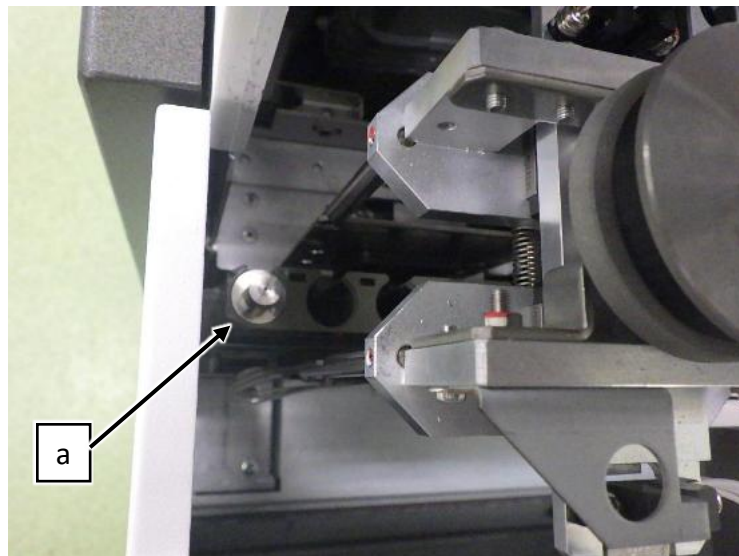
(b): SAMPLE PROBE ADJUSTING TOOL 2 (P/N:792-6427)

- (7) Adjust Sample probe position so that the tip of sample probe is at the center hole of the tool using the 4 arrow buttons.

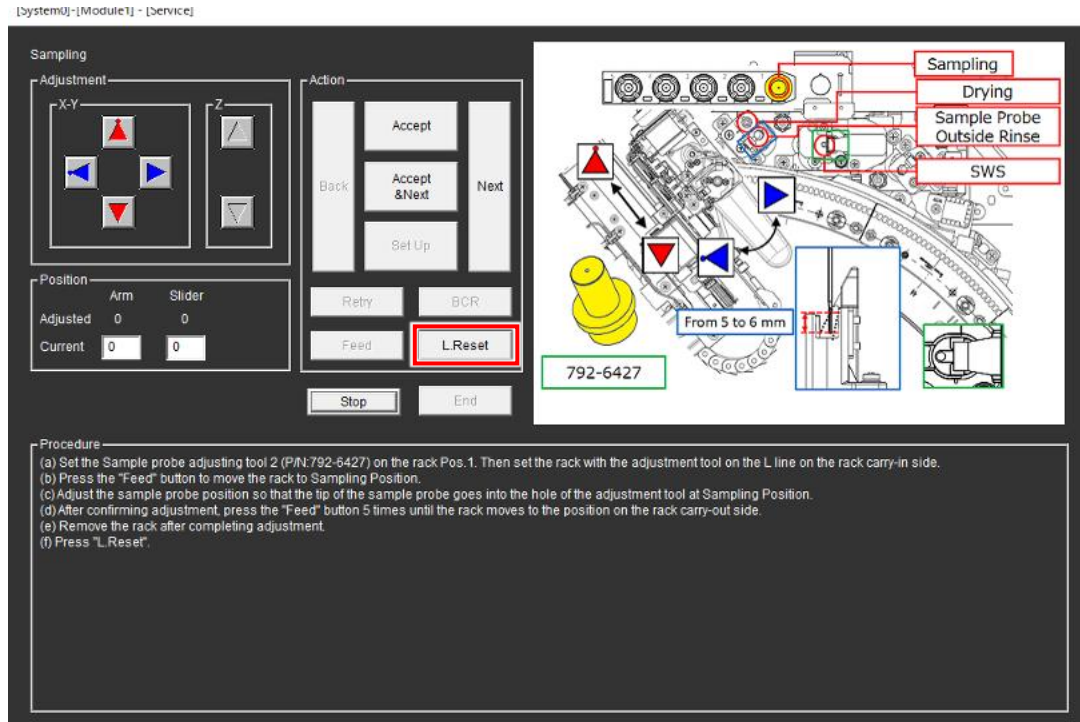
(Range: arm: ± 80 pulse, slider: ± 160 pulse)



- (8) Select [Accept].
- (9) Select [Feed] 5 times.
- (10) Remove the rack (a) on the left end of SBL.



(11) Select [L.Reset].

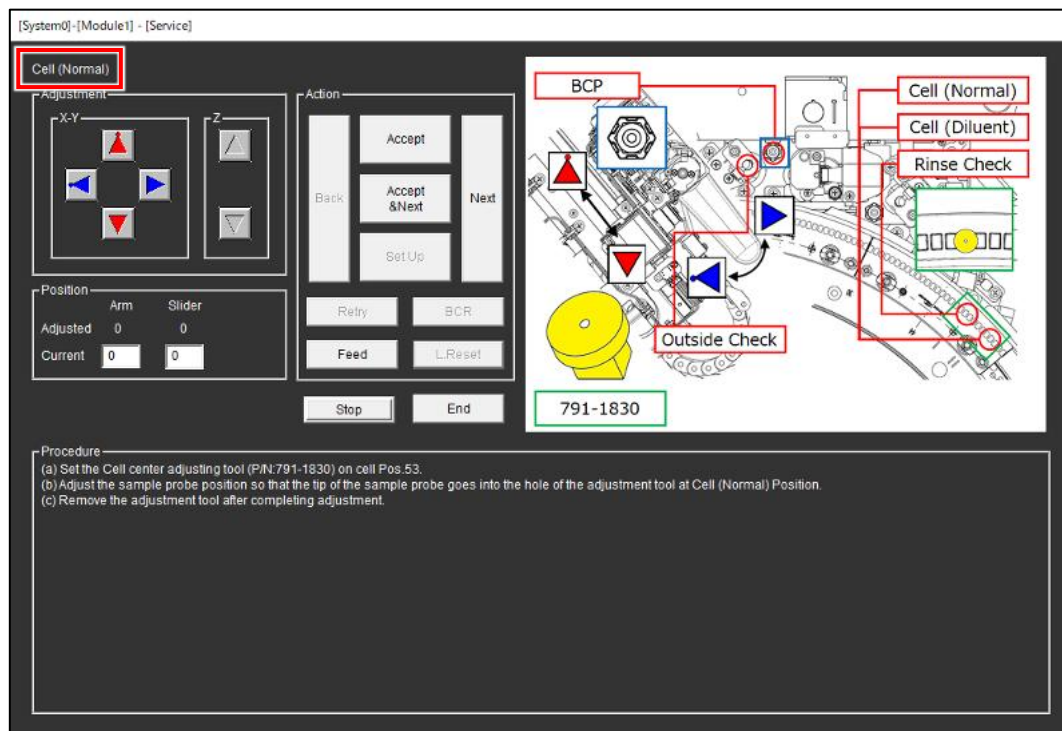


(12) For next adjustment, select [Next].

3.3 Cell (Normal) position

(1) Make sure that the adjustment subject is "Cell (Normal)" at the upper left of SSW.

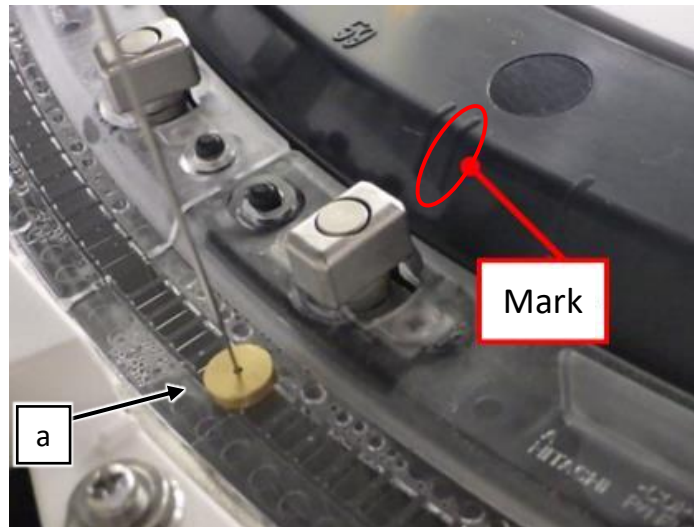
If not, select [Next] repeatedly until "Cell (Normal)" appears.



- (2) Set the tool (a) on No.53 cell according to the mark (right line).
- (3) Lower Sample probe by rotating the pulley and make sure that the tip of Sample probe is at the center of the tool (a).

If it is at the center, select [Next] and proceed to the next adjustment.

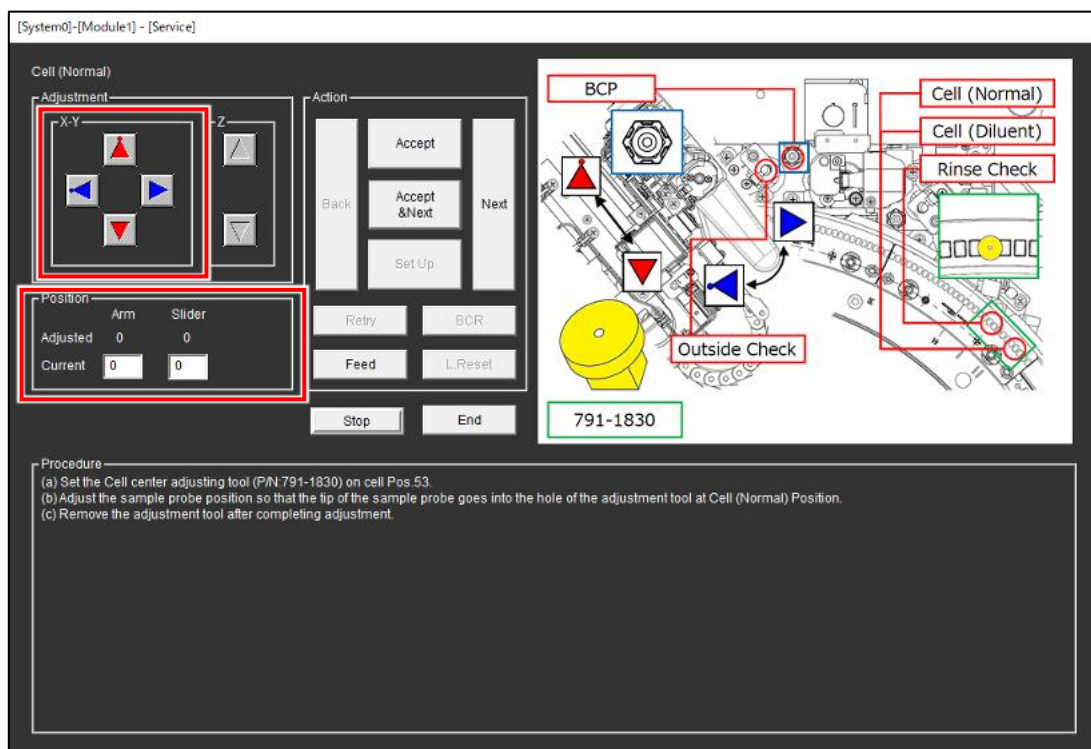
If not, proceed to the next step.



(a): CELL CENTER ADJUSTING TOOL (P/N:791-1830)

- (4) Adjust Sample probe position so that the tip of Sample probe is at the center hole of the tool using the 4 arrow buttons.

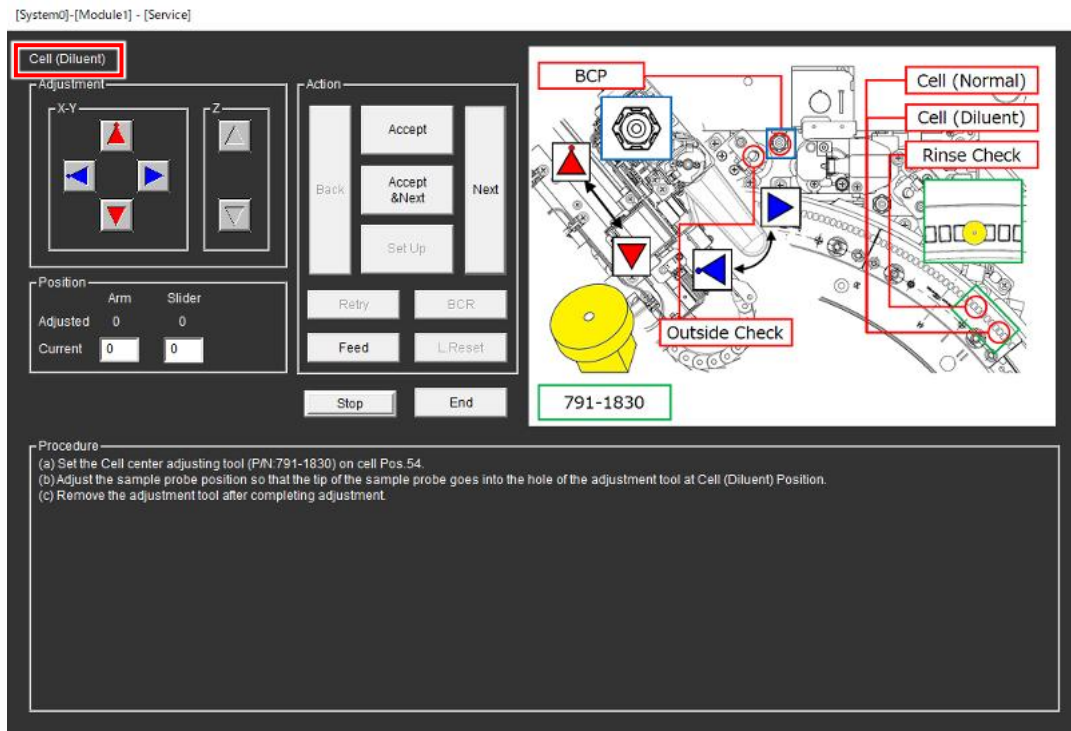
(Range: Arm: ± 80 pulse, Slider: ± 160 pulse)



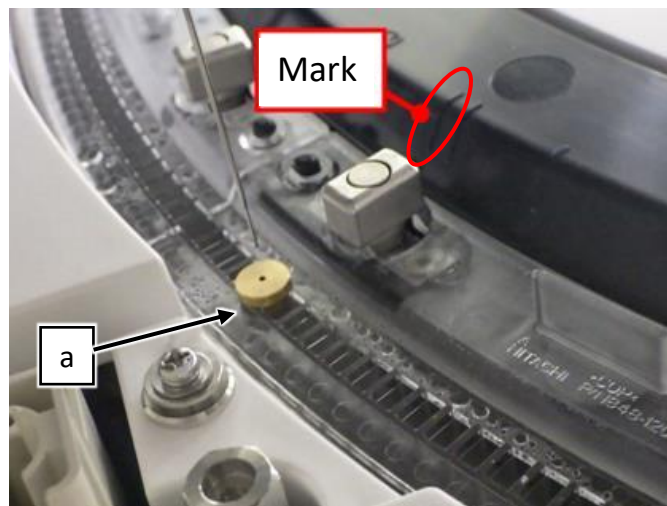
- (5) Remove the tool from No.53 cell.
- (6) Select [Accept & Next].

3.4 Cell (Dilution) position

- (1) Make sure that the adjustment subject is “Cell (Dilution)” at the upper left of SSW.
If not, select [Next] repeatedly until “Cell (Dilution)” appears.

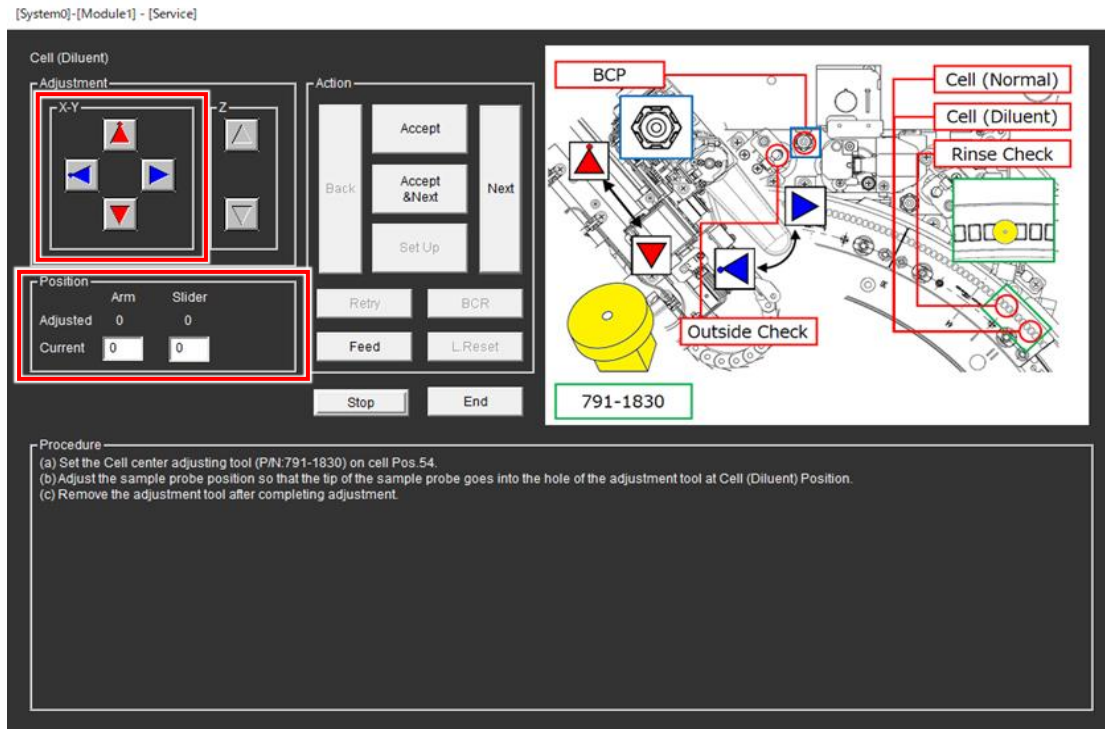


- (2) Set the tool (a) on No.54 cell, according to the mark (left line).



(a): CELL CENTER ADJUSTING TOOL (P/N:791-1830)

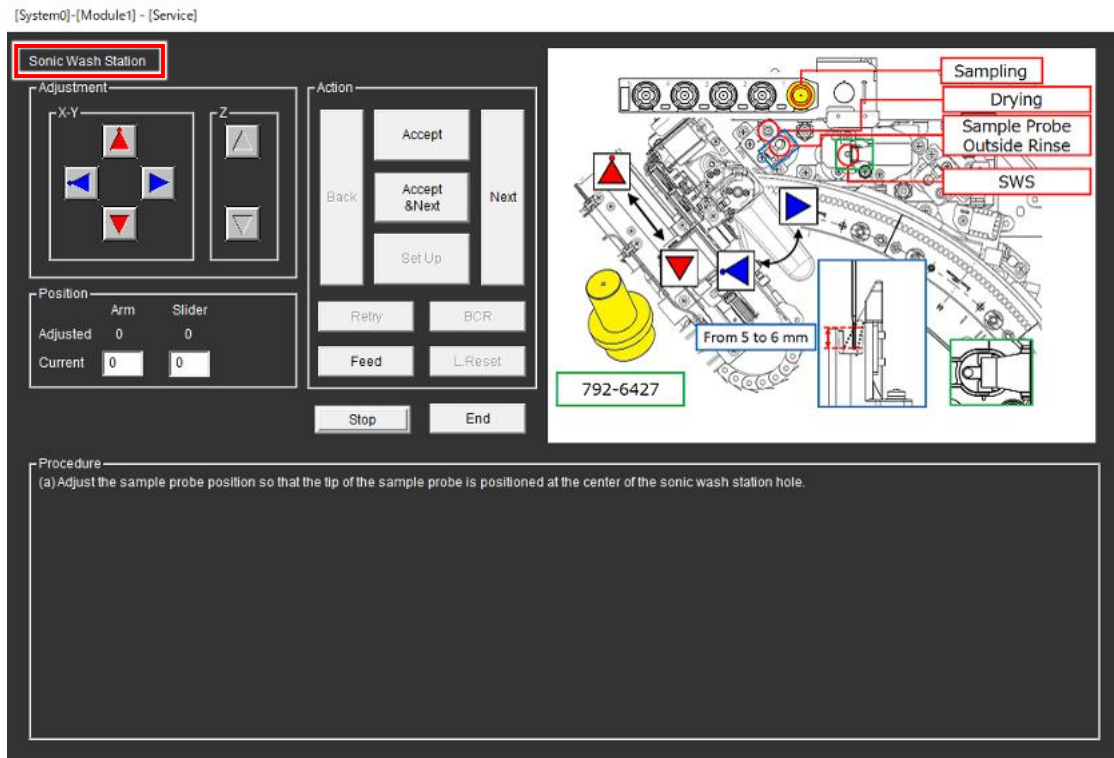
- (3) Lower Sample probe by rotating the pulley and make sure that the tip of Sample probe is at the center of the tool.
If it is at the center, select [Next] and proceed to the next adjustment.
If not, proceed to the next step.
- (4) Adjust Sample probe position so that the tip of Sample probe is at the center hole of the tool using the 4 arrow buttons.
(Range: Arm: ± 80 pulse, Slider: ± 160 pulse)



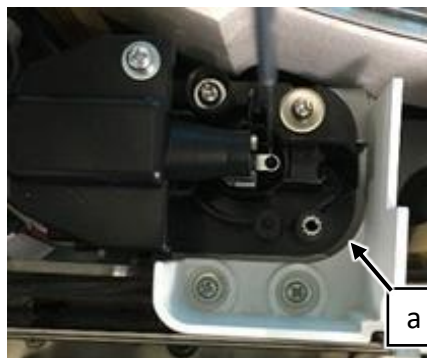
- (5) Remove the tool from No.54 cell.
- (6) Select [Accept & Next].

3.5 Sonic Wash Station position

- (1) Make sure that the adjustment subject is “Sonic Wash Station” at the upper left of SSW.
If not, select [Next] repeatedly until “Sonic Wash Station” appears.

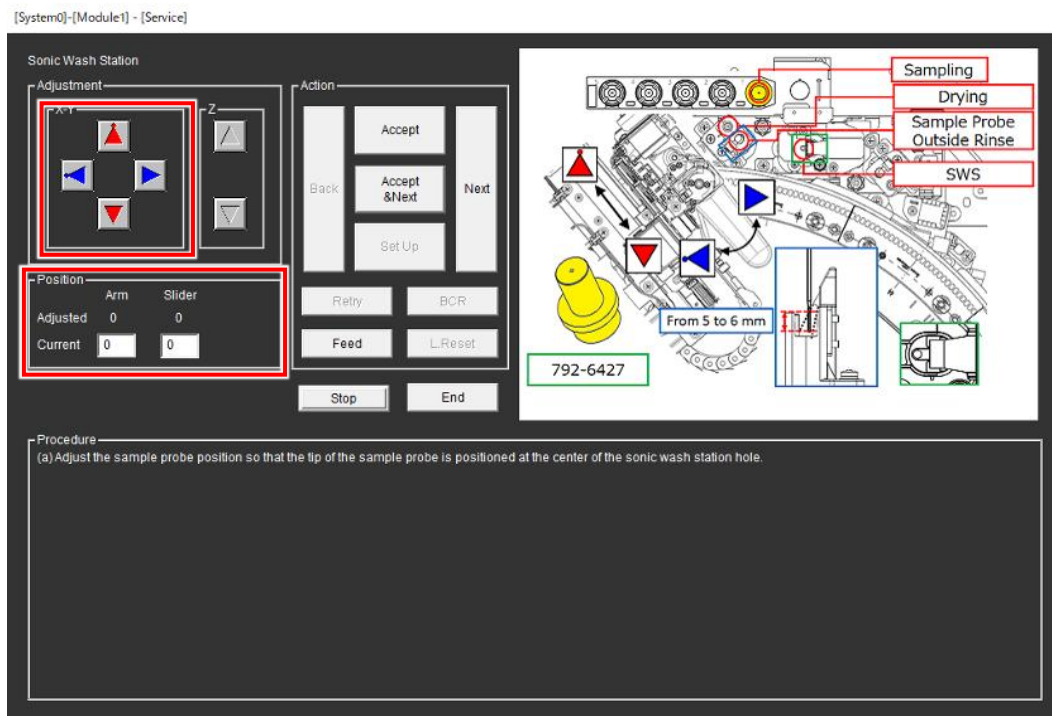


- (2) Lower Sample probe by rotating the pulley and make sure that the tip of Sample probe is at the center of hole of the sample washing station.
If it is at the center, select [Next] and proceed to the next adjustment.
If not, proceed to the next step.



- (3) Adjust Sample probe position so that the tip of Sample probe is at the center of the hole using the 4 arrow buttons.

(Range: Arm: ± 80 pulse, Slider: ± 160 pulse)

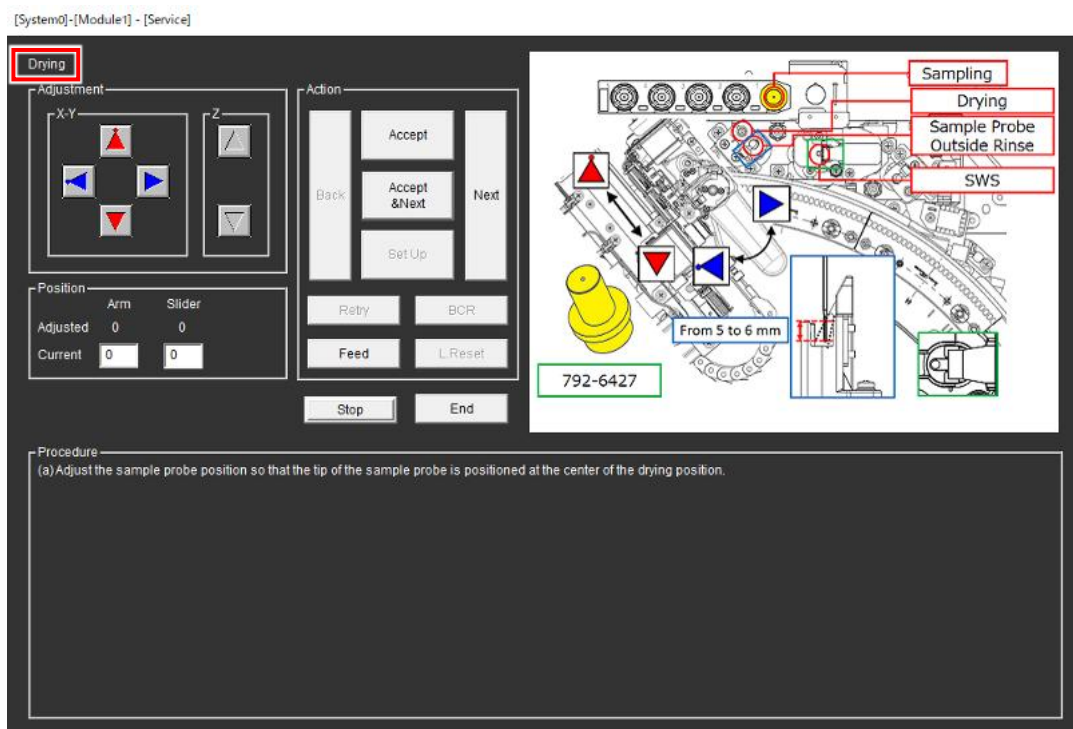


- (4) Select [Accept & Next].

3.6 Drying position

- (1) Make sure that the adjustment subject is "Drying" at the upper left of SSW.

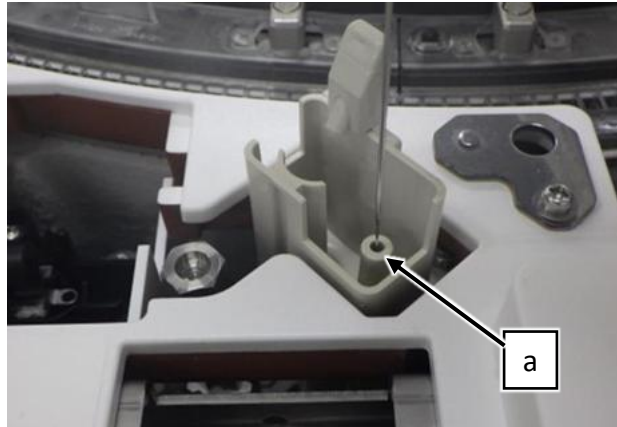
If not, select [Next] repeatedly until "Drying" appears.



- (2) Lower Sample probe by rotating the pulley and make sure that the tip of Sample probe is at the center of the drying hole (a) of the washing station.

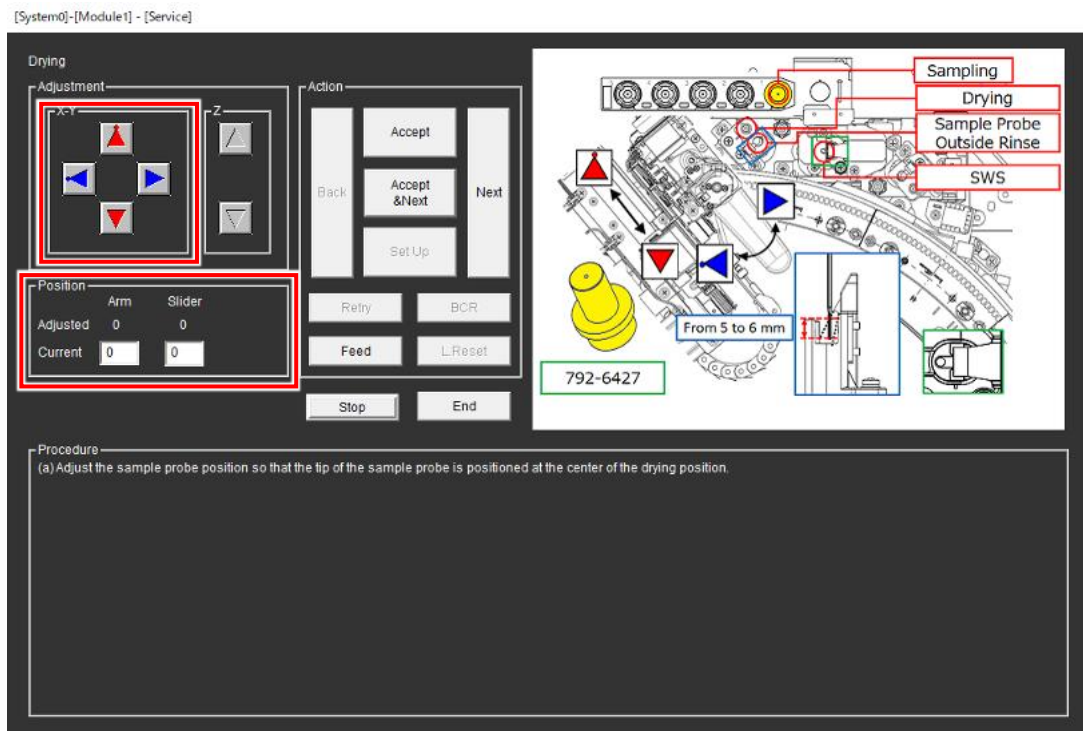
If it is at the center, select [Next] and proceed to the next adjustment.

If not, proceed to the next step.



- (3) Adjust Sample probe position so that the tip of Sample probe is at the center of the drying hole using the 4 arrow buttons.

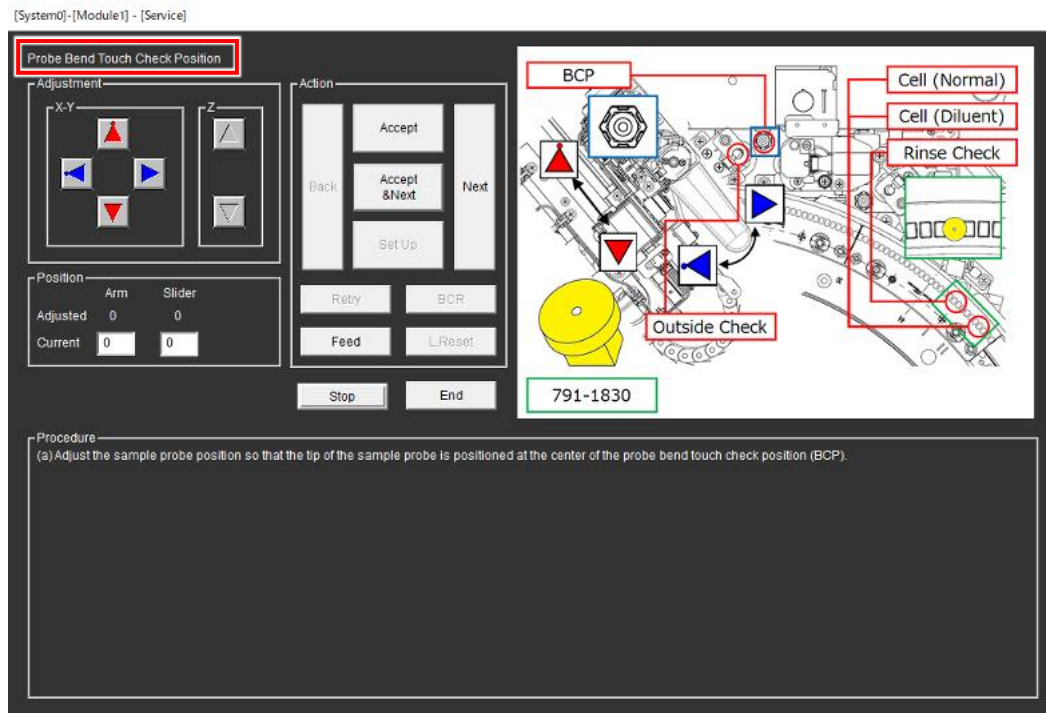
(Range: Arm: ± 80 pulse, Slider: ± 160 pulse)



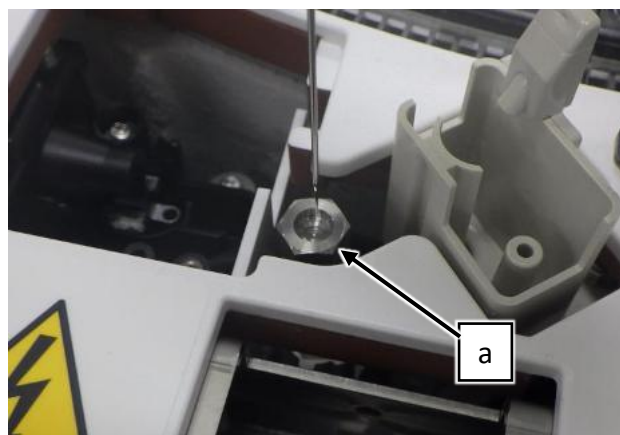
- (4) Select [Accept & Next].

3.7 Probe Bend Touch Check Position

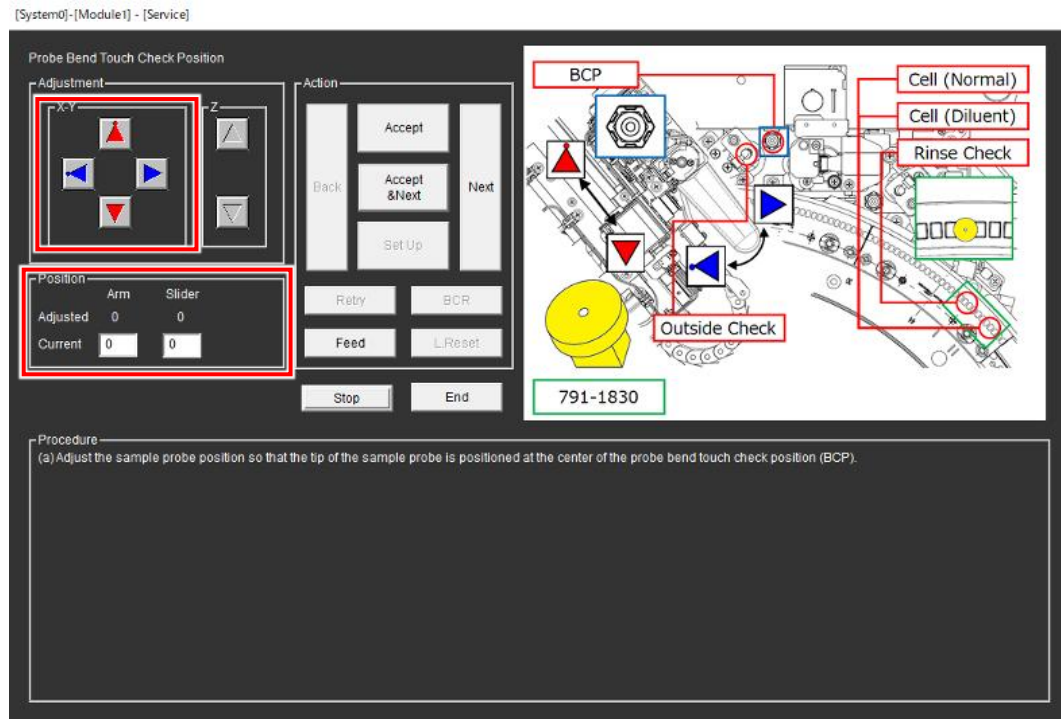
- (1) Make sure that the adjustment subject is “Probe Bend Touch Check Position” at the upper left of SSW.



- (2) If not, select [Next] repeatedly until “Probe Bend Touch Check Position” appears.
- (3) Lower Sample probe by rotating the pulley and make sure that the tip of Sample probe is at the center of the hole (a).
If it is at the center, select [Next] and proceed to the next adjustment.
If not, proceed to the next step.



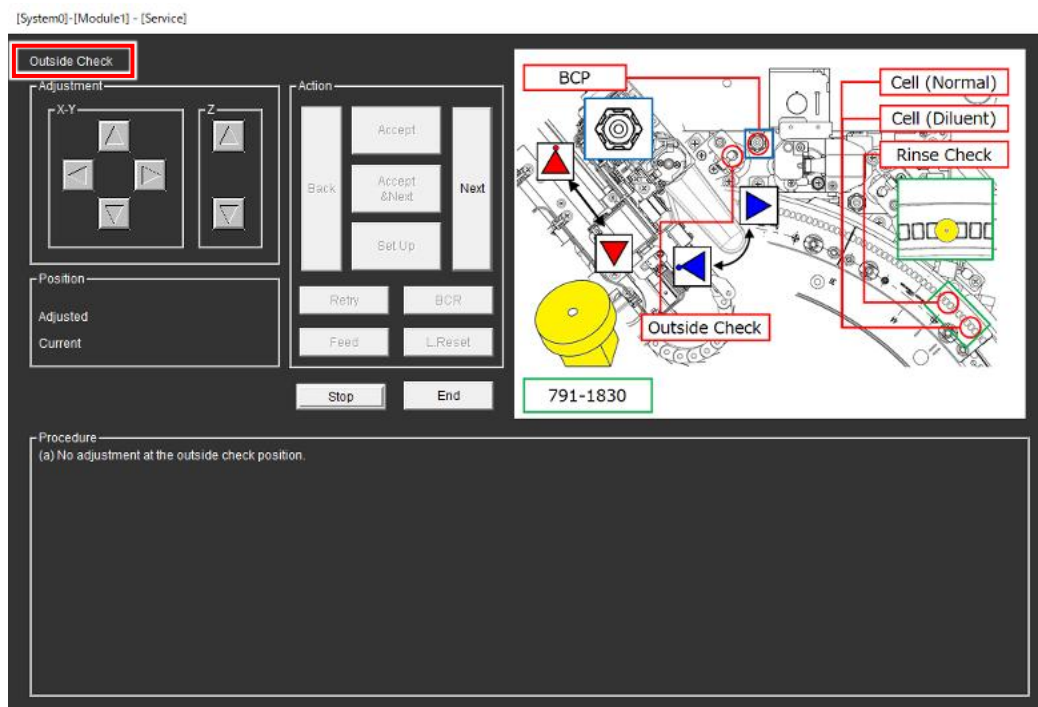
- (4) Adjust Sample probe position so that the tip of Sample probe is at the center of the hole using the 4 arrow buttons.
(Range: Arm: ± 80 pulse, Slider: ± 160 pulse)



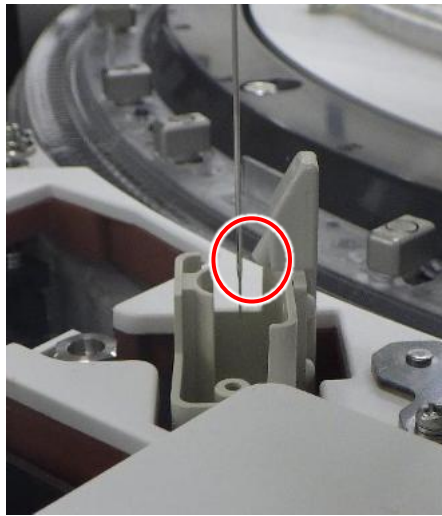
- (5) Select [Accept & Next].

3.8 Outside Check position

- (1) Make sure that the adjustment subject is "Outside Check" at the upper left of SSW.
If not, select [Next] repeatedly until "Outside Check" appears.



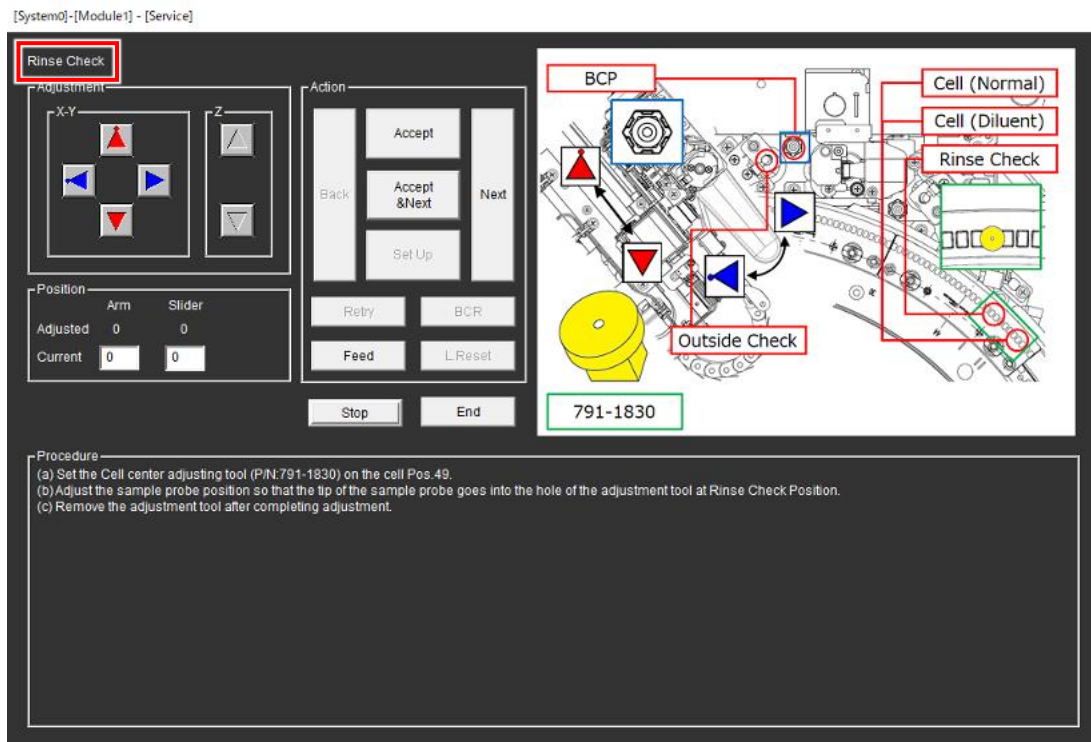
- (2) Make sure that the tip of the sample probe does not touch the outside washing outlet of the washing station.
If not, perform Sample Probe Wash Volume Adjustment (refer to “S03-09 Sample Probe Wash Volume Adjustment”).



- (3) Select [Next].

3.9 Rinse Check position (Sample Probe A only)

- (1) Make sure that the adjustment subject is “Rinse Check” at the upper left of SSW.
If not, select [Next] repeatedly until “Rinse Check” appears.

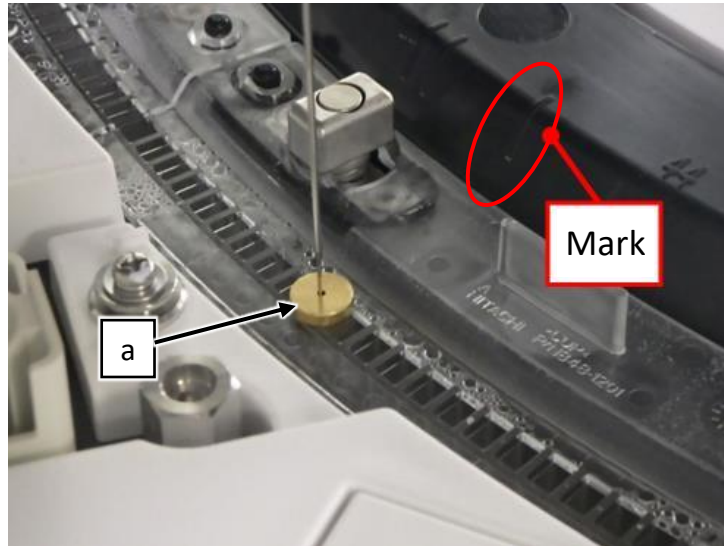


- (2) Set the tool (a) on No.49 cell, according to the mark.

- (3) Lower Sample probe by rotating the pulley and make sure that the tip of Sample probe is at the center of the tool (a).

If it is at the center, select [Next] and proceed to the next adjustment.

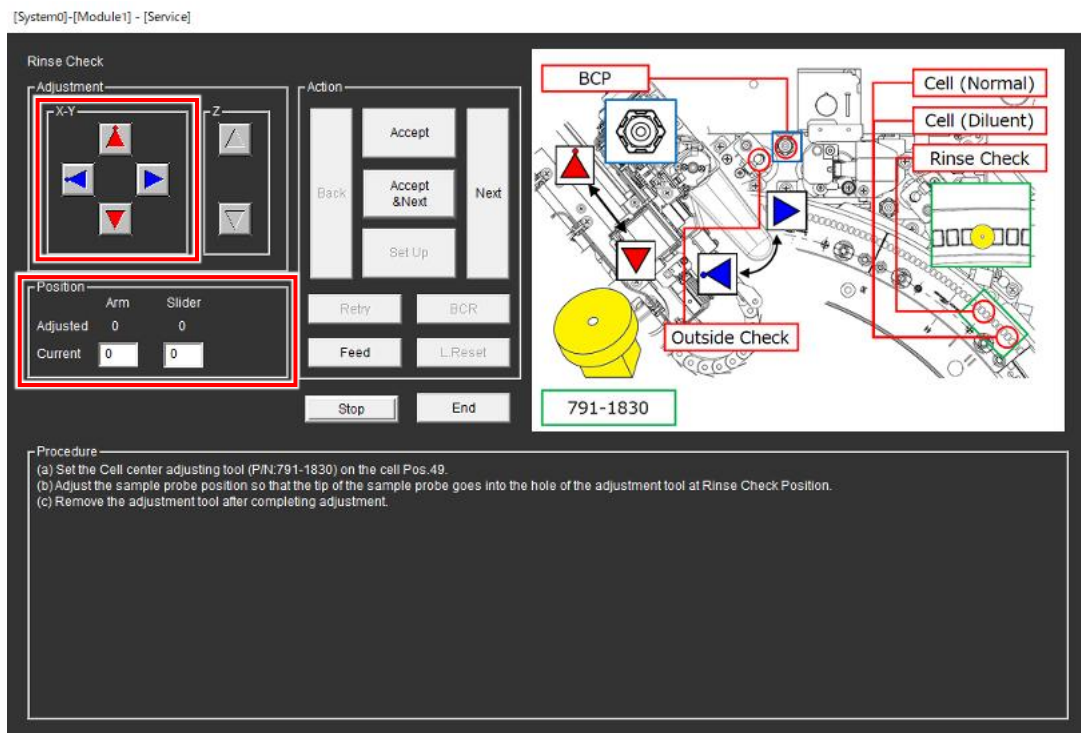
If not, proceed to the next step.



(a): CELL CENTER ADJUSTING TOOL (P/N:791-1830)

- (4) Adjust Sample probe position so that the tip of Sample probe is at the center hole of the tool using the 4 arrow buttons.

(Range: Arm: ± 80 pulse, Slider: ± 160 pulse)



- (5) Remove the tool from No.49 cell.
- (6) Select [Accept & Next].
- (7) Make sure no racks remain on Sampling line.
- (8) Select [End].

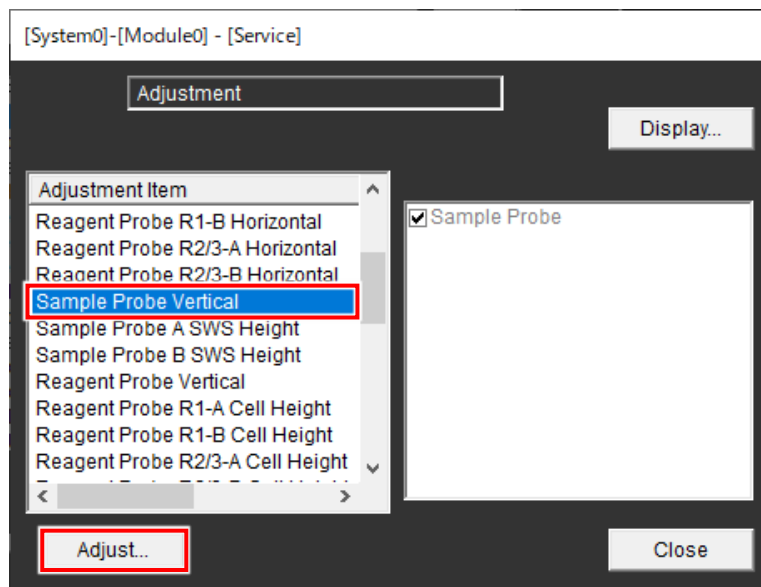
4. Adjustment of Sample Probe B Horizontal

The adjustment procedure for SB is basically same as the procedure for SA.

Perform Sample Probe B adjustment in the same steps (refer to the section 3.1 to 3.8).

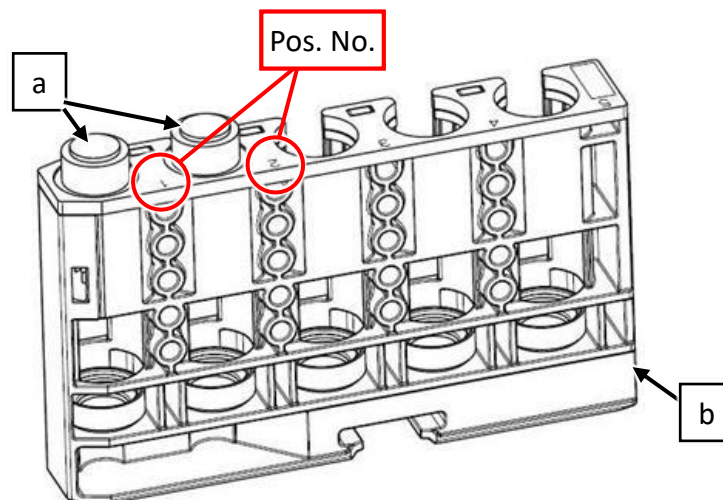
5. Adjustment of Sample Probe Vertical

- (1) Select [Sample Probe Vertical] from the Adjustment item and select [Adjust...] in SSW.



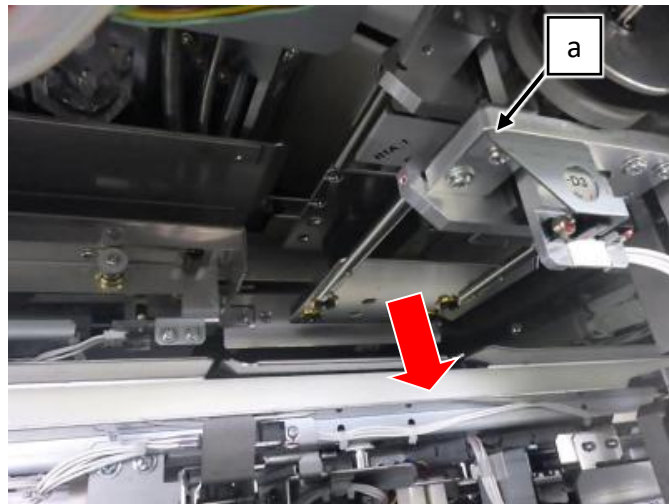
- (2) Set the 2 tools (a) on Pos.1 and Pos.2 of the rack (b).

Note: Be careful of the direction of the tools (a).

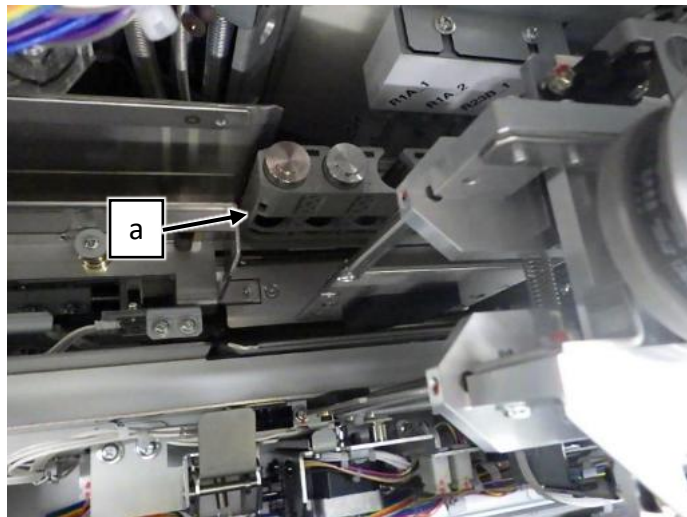


(a): SAMPLE PROBE ADJUSTING TOOL 2 (P/N:792-6427)

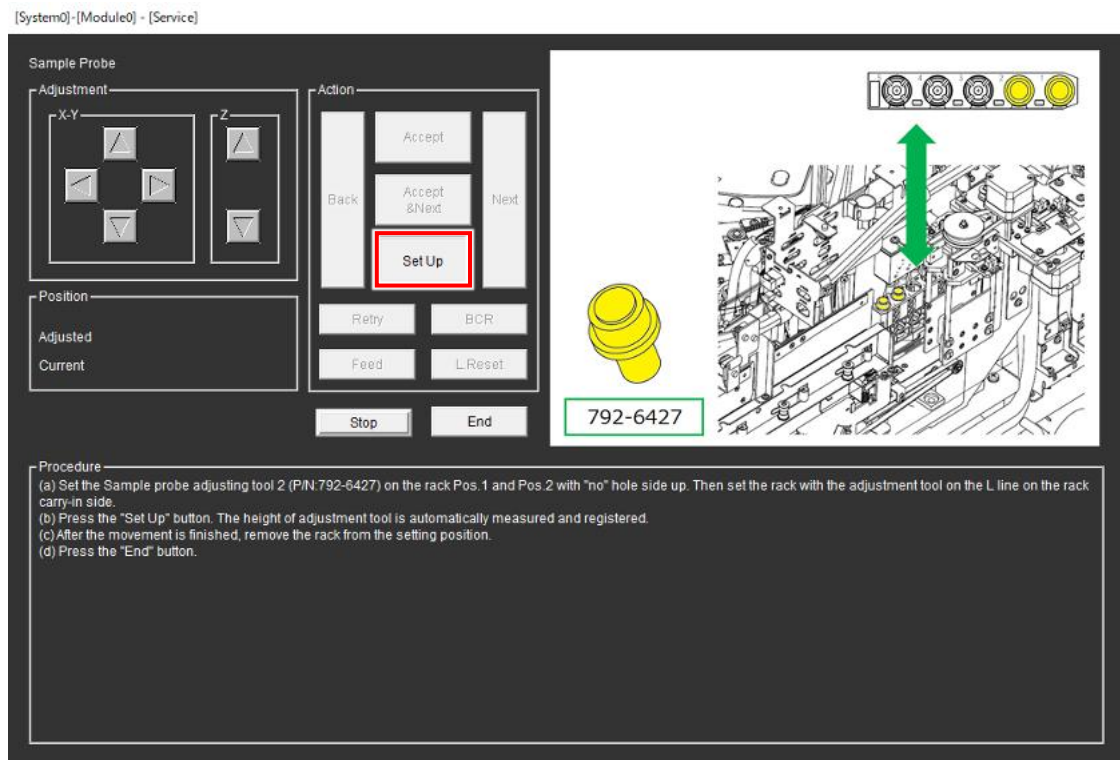
- (3) Move Rack handle (a) as shown below.



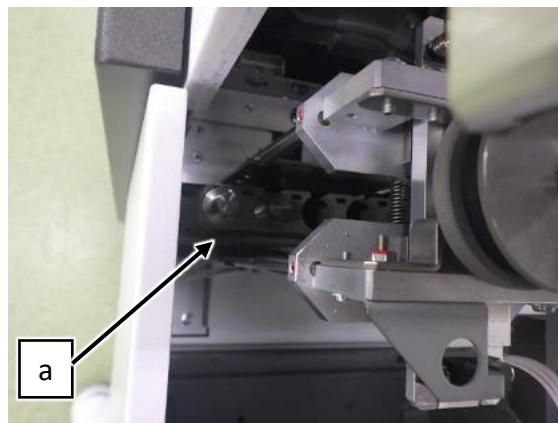
- (4) Set the rack (a) on the right end of Sample line.



- (5) Select [Set Up].



- (6) Wait for the automated adjustment of Sample Probe Vertical to end.
- (7) Make sure that any alarm does not occur in SSW and SSU.
- (8) Remove the rack (a) on the right end of SBL.

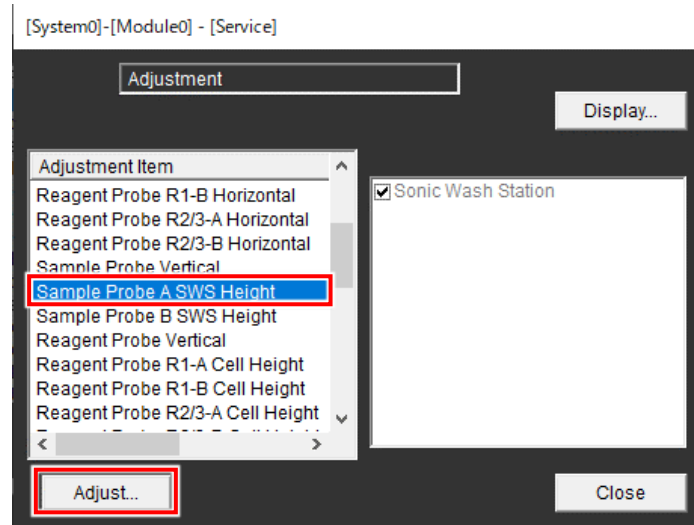


- (9) Select [End].

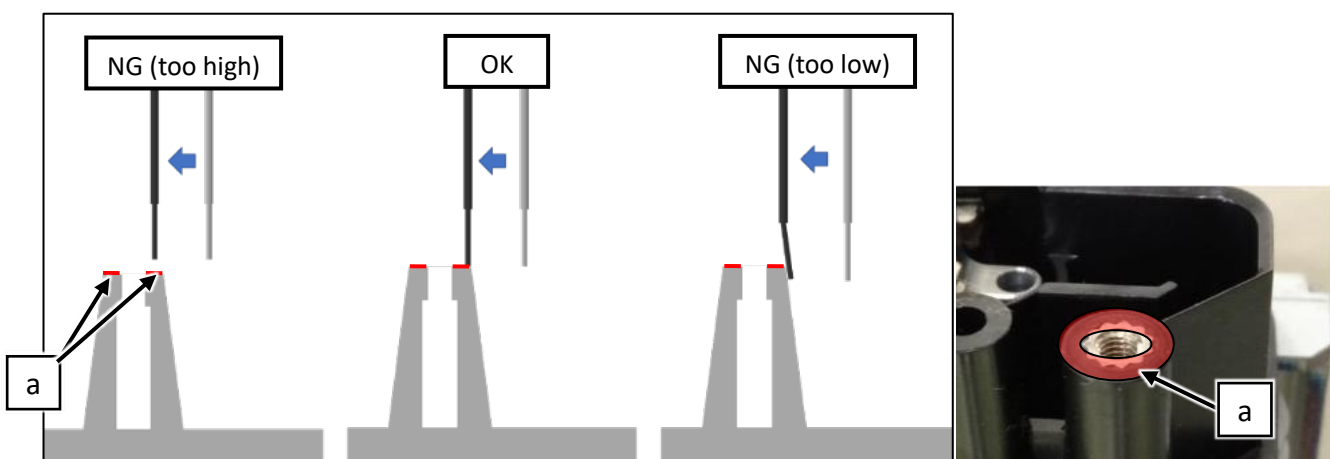
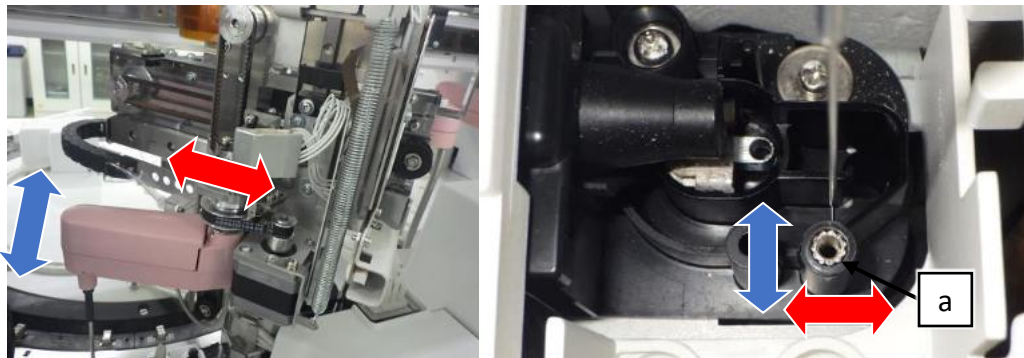
6. Adjustment of Sample Probe A SWS Height

The adjustment procedure for sample probe B is basically same as the procedure for sample probe A.

- (1) Select [Sample Probe A SWS Height] from Adjustment item and select [Adjust...] in SSW.

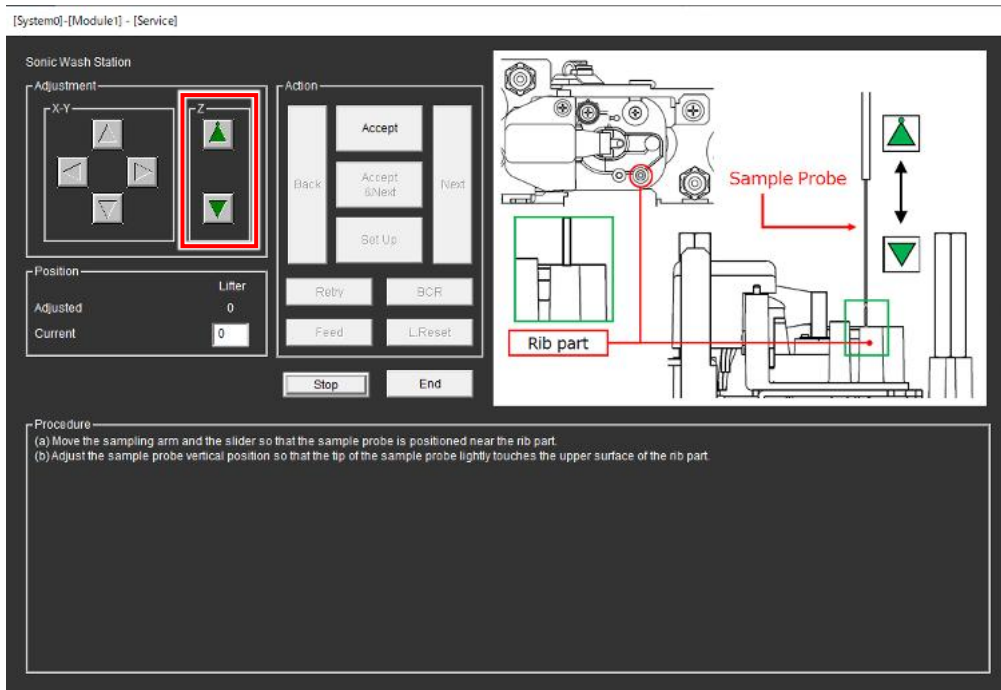


- (2) Move Sample probe and make sure that the tip of sample probe touches the target area (a) lightly.



- (3) Adjust the height of Sample probe so that the tip of sample probe touches the target area lightly using the 2 arrow buttons.

(Range: -107 pulse - 213 pulse)



- (4) Select [Accept].
(5) Select [End].

7. Adjustment of Sample Probe B SWS Height

The adjustment procedure for SB is basically same as the procedure for SA.

Perform Sample Probe B SWS Height adjustment in the same steps (refer to the chapter 6).

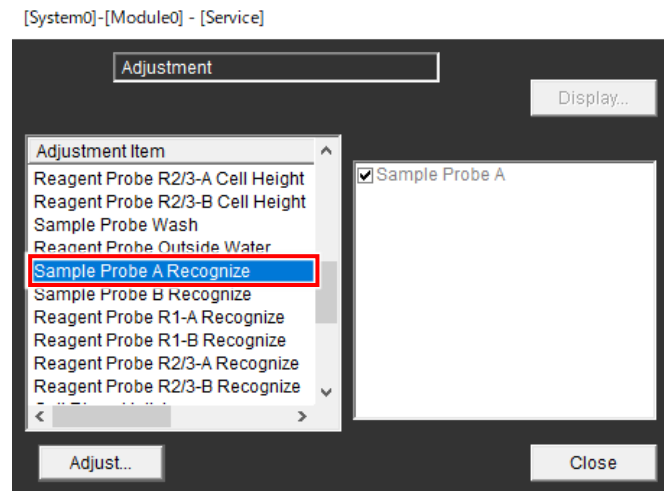
8. Adjustment of Sample Probe A Recognize

The adjustment procedure for sample probe B is basically same as the procedure for sample probe A.

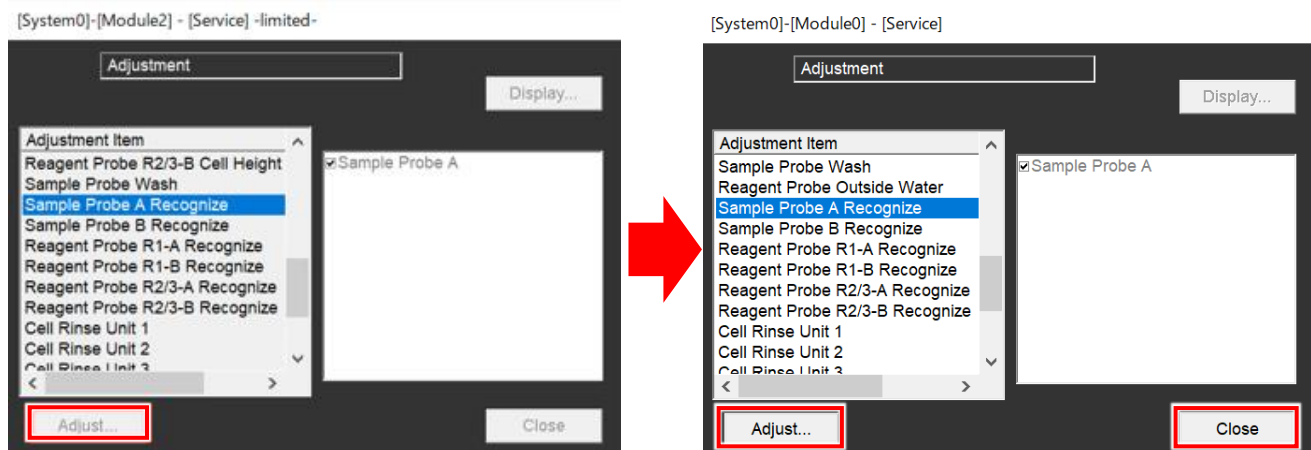
Make sure the chapter 3 - 7 have been completed.

If they haven't been completed yet, perform the adjustments.

- (1) Select [Sample Probe A Recognize] from the Adjustment item and select [Adjust...] in SSW.



- (2) Wait for the automated adjustment of Bend Check position to end.
Each Sample Probe Recognize takes approximately 2 minutes.
- (3) If alarm is occurred, perform the section 3.7 and the chapter 5.
- (4) Make sure that [Adjust] is active, and then select [Close].



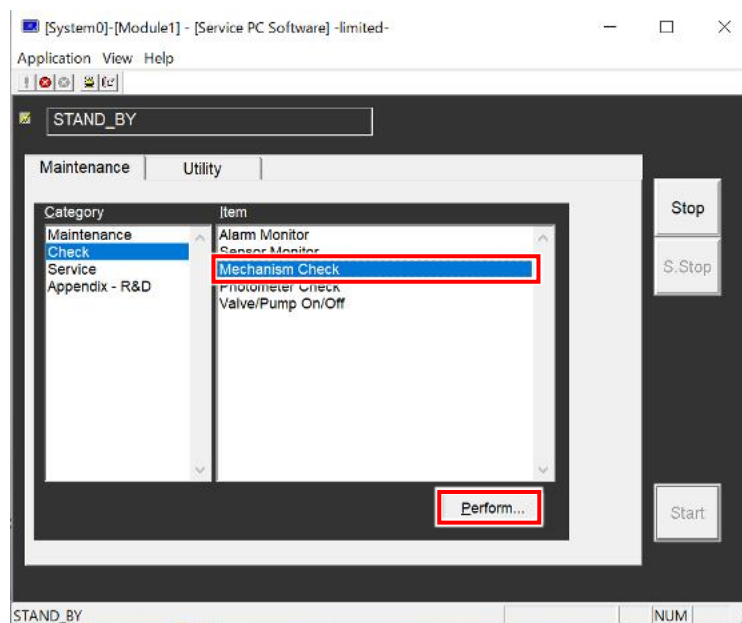
9. Adjustment of Sample Probe B Recognize

The adjustment procedure for SB is basically same as the procedure for SA.

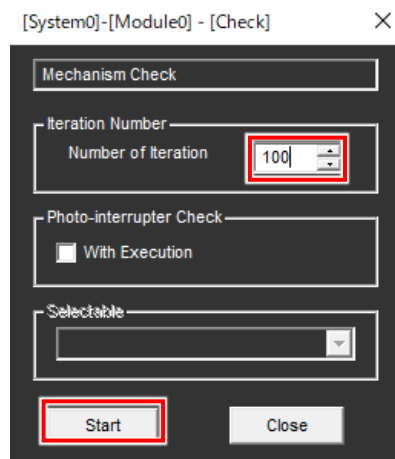
Perform Sample Probe B Recognize adjustment in the same steps (refer to the chapter 8).

10. Accomplishing the adjustment

- (1) Attach No.24, No.27, and No.28 covers.
- (2) Attach COVER SWS T.
- (3) Attach SWS COVER U ASSY to TB COVER SSP ASSY.
- (4) Attach Cell cover.
- (5) Close TOP COVER R ASSY.
- (6) Select [Check] > [Mechanism Check] > [Perform...] in SSW.



- (7) Set the number of iterations to 100 and select [Start].



- (8) Wait for Mechanism Check to end.
It takes approximately 3 minutes.
- (9) Make sure that adjusted Sample probe do not collide anything.
If it collides something, retry the adjustments of colliding Sample probe.
- (10) Turn off Maintenance switch.
- (11) Attach COVER MSW ASSY.

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